Processes, Threads and Context Switching

LumoraAI Presentation Demo • 02 Operating Systems

Processes

• A process is an executing program with its own virtual address space and operating-system-managed resources. Process isolation prevents an ordinary process from directly accessing another process's memory.

• A process control block stores operating-system state such as process state, scheduling information and register-related context.

Threads

• Threads are execution units within a process. Threads of the same process normally share code, data and heap memory while maintaining separate stacks and execution state.

Context switching

• A context switch saves the state of the currently running execution unit and restores the state of another. Switching introduces overhead because useful application work pauses while the operating system changes execution context.

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